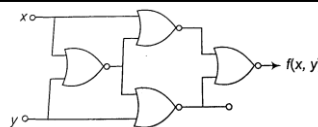




COMPUTER AWARENESS

- A CPU has an arithmetic unit that adds bytes and then sets its V, C and Z flag bits as follows: the V-bit is set if arithmetic overflow occurs. The C-bit is set if a carry-out is generated from the most significant bit during an operation. The Z-bit is set if the result is zero. What are the values of the V, C and Z flag bits respectively after the 8-bit bytes 1100 1100 and 1000 1111 are added?
(a) 0, 0, 0 (b) 1, 1, 0
(c) 1, 1, 1 (d) 0, 1, 0
- Which of the following statements is always true?
(a) A compiled program uses more memory than an interpreted program.
(b) A compiler converts a program to a lower level language for execution.
(c) A compiler for a high level language takes less memory than its interpreter
(d) Compiled programs take more time to execute than interpreted programs.
- Floating point numbers in a computer are represented using a 10-bit mantissa (including a sign bit) and a 7-bit exponent (including a sign bit). What is the approximate value of the maximum number, which can be represented? Assume that the mantissa is stored in the normalised form, that is, without leading zeroes.
(a) 2^{128} (b) 2^{127} (c) 2^{64} (d) 2^{63}
- The capacity of a memory unit is defined by the number of words multiplied by the number of bits per word. How many separate address and data line are needed for a memory of $4K \times 16$?
(a) 10 address lines and 16 data lines
(b) 12 address lines and 10 data lines
(c) 12 address lines and 16 data lines
(d) 12 address lines and 8 data lines
- The main disadvantage of direct mapping of cache organization is that
(a) it doesn't allow simultaneous access to the intended data and its tag
(b) it is more expensive than other type of organizations.
(c) the cache hit ratio is degraded if two or more blocks used alternatively map onto the same block frame in the cache.
(d) the number of blocks required for the cache increases linearly with the size of the main memory.
- Let $A[1 \dots 10]$ be an array, let $A[i] = 2i$ for $1 \leq i \leq 10$. After the assignment $j = A[A[5]]$ is executed, the value of $A[j]$ is equal to
(a) undefined (b) 1
(c) 5 (d) 10
- The first instruction of bootstrap loader program of an operating system is stored in
(a) RAM (b) BIOS
(c) Hard disk (d) None of these
- The function $AB'C + AB'C + ABC' + A'B'C$ is equivalent to
(a) $AC' + AB + A'C$ (b) $AB' + AC' + A'C$
(c) $A'B + AC' + AB'$ (d) $A'B + AC + AB'$
- The addition of 4 bit, 2's complement binary numbers 1101 and 0100 results is
(a) 0001 and an overflow (b) 1001 and no overflow
(c) 0001 and no overflow (d) 1001 and an overflow
- Given $\sqrt{(224)}_r = (13)_r$, the value of radix r is
(a) 10 (b) 8 (c) 6 (d) 5
- Let $A = 11111010$ and $B = 00001010$ be two 8 bit 2's complement number. Their product in 2's complement is
(a) 11000100 (b) 10011100
(c) 10100101 (d) 11010101
- Identify the logic function performed by the circuit



- (a) Exclusive OR (b) Exclusive NOR
(c) NAND (d) NOR

- Which of the following is (are) true about virtual memory system that uses pages?
(I) The virtual address space can be larger than the amount of physical memory.
(II) Programs must be resident in main memory throughout their execution.
(III) Pages correspond to semantic characteristics of the programs.
(a) I only (b) II only
(c) I and II (d) I and III
- The minimum number of gates needed to implement the Boolean function $f(x, y, z) = z(x + y) + (\bar{z} + x + y)(\bar{x} + \bar{y})$ is
(a) 2 (b) 3 (c) 4 (d) 5
- How many bits are required to store an ASCII character?
(a) 7 (b) 6 (c) 8 (d) None

ANSWER

1.	B	2.	B	3.	D	4.	C	5.	B
6.	A	7.	B	8.	B	9.	C	10.	D
11.	A	12.	B	13.	C	14.	A	15.	a

SOLUTION

- (b)
 $1 \rightarrow \text{carry}$

$$\begin{array}{r} 11001100 \\ + 10001111 \\ \hline 101011011 \end{array}$$

$V = 1$, as there is an overflow
 $C = 1$, as carry out is generated by most significant bits
 $Z = 0$, as the result is not zero.
Hence, 1,1,0 is the bits obtained.
- (b)
A compiler converts a high level program into low level language (Machine language) for execution.
- (d)

↓ Sign bit

--	--	--	--	--	--	--	--

Mantissa

↓ Sign bit

--	--	--	--	--	--	--	--

Exponent

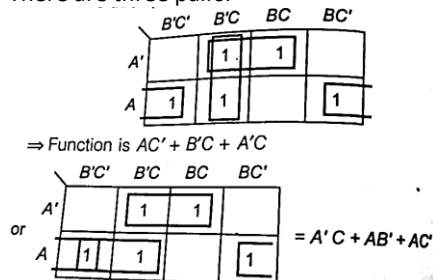
Maximum number
 $= 11111111 \times 2^{(1111111)_2} = 2^{63}$
- (c) $4K \times 16 = 2^{12} \times 16$
 $\Rightarrow$ We should have 12 address lines and 16 data lines.
- (d)
The number of blocks required for the cache increases linearly with the size of the main memory which is the main disadvantage of direct mapping of cache organization.
- (a) Given, $A[i] = 2i$, where $1 \leq i \leq 10$
 $\Rightarrow A[10] = 2 \times 10 = 20$
Now, $j = A[A[5]] = A[2 \times 5] = A[10] \Rightarrow j = 20$
 $\Rightarrow A[j] = A[20]$





Which is not defined because A takes the values from 1 to 10. But here, the condition occurs of overflow.

7. (b) BIOS = Basic input Output System.
8. (b) It will be simplified by Karnaugh map as follows. There are three pairs.

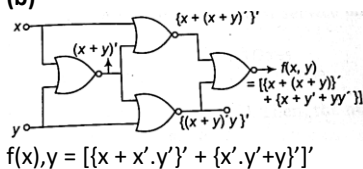


9. (c)
1's is complement of $1101 = 0010$
2's complement of $1101 = 0010 + 1 = 0011$
And 1's complement of $0100 = 1011$
2's complement of $0100 = 1011 + 1 = 1100$
Addition both 4 bits
- $$\begin{array}{r} 0011 \\ + 1100 \\ \hline 1111 \end{array}$$
- It's one's complement = 0000
And 2's complement = $0000 + 1 = 0001$
There is no overflow.

10. (d) $\sqrt{(224)_r} = (13)_r \Rightarrow \sqrt{2r^2 + 2r + 4} = r + 3$
 $\Rightarrow 2r^2 + 2r + 4 = (r + 3)^2 = r^2 + 6r + 9$
 $\Rightarrow r^2 - 4r - 5 = 0$
 $\Rightarrow (r - 5)(r + 1) = 0$
 $\Rightarrow r = -1, 5$
As a cannot be -1 , so $r = 5$.

11. (a)
2's complement of $A = 00000110$
2's complement of $B = 11110110$
 $11110110 = (246)_{10}$
 $\frac{\times 00000110}{00000000} = (6)_{10}$
 11110110
 11110110
 $10111000100 = (1476)_{10}$
8, bit representation of, 10111000100 is
 $11000100 = (196)_{10}$

12. (b)



$$f(x, y) = [(x + x')y]' + [(x + y')y']'$$

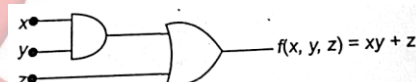
$$\begin{aligned} &= [x' \cdot (x')' \cdot y]' + [x' \cdot y']' && \text{(by De Morgan's law)} \\ &= [x' \cdot (x + y) + (x + y) \cdot y']' && \text{(By De Morgan's law)} \\ &= [x' \cdot (x + y)]' \cdot [(x + y) \cdot y']' && \text{(By De Morgan's law)} \\ &= [x + (x + y)'] \cdot [(x + y)' + y] && \text{(By De Morgan's law)} \\ &= [x + x' \cdot y'] \cdot [x' + y] && \text{(By De Morgan's law)} \\ &= (x + x') \cdot (x + y') \cdot (x' + y) \cdot (y' + y) && \text{(By De Morgan's law)} \\ &= 1 \cdot (x + y') \cdot (x' + y) \cdot 1 && \text{(By complement law)} \\ &= (x + y') \cdot (x' + y) && \text{(By identity law)} \\ &= x \cdot x' + xy + y'x' + y'y && \text{(By Distributive law)} \\ &= 0 + xy + y'x' + 0 && \text{(By complement law)} \\ &= xy + x'y' && \text{(By identity and commutative law)} \end{aligned}$$

Which represent exclusive NOR.

13. (c)

14. (a)

$$\begin{aligned} f(x, y, z) &= z(x + y) + (\bar{z} + x + y)(\bar{x} + \bar{y}) \text{ (given)} \\ &= z(x + y) + (z\bar{x}\bar{y}) + (xy) && \text{(By De Morgan's Law)} \\ &= xy + yz + zx + z\bar{x}\bar{y} && \text{(By Distributive law)} \\ &= yz + zx + (xy + z\bar{x}\bar{y}) && \text{(By Distributive law)} \\ &= yz + zx + (xy + z)(\bar{x} + \bar{y}) && \text{(By complement law)} \\ &= yz + zx + (xy + z) \cdot 1 && \text{(By complement law)} \\ &= yz + zx + xy + z && \text{(By Absorption law)} \\ &= zx + xy + z \cdot 1 && \text{(By Absorption law)} \\ &= xy + z(x + 1) = xy + z \cdot 1 = xy + z \\ &\Rightarrow f(x, y, z) = xy + z \end{aligned}$$



It requires two gates.

15. (a)